

ASSIGNMENT ---1.5

Q1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Components: --

1. **Motherboard** – The main circuit board that connects all the computer parts together.
2. **CPU (Processor)** – The brain of the computer. It processes all instructions and calculations.
3. **RAM (Memory)** – Temporary memory used while programs are running. It helps the computer work faster.
4. **HDD (Hard Disk Drive)** – Stores the operating system, software, and files permanently.
5. **SSD (Solid State Drive)** – A faster storage device that helps the computer boot and load programs quickly.
6. **SMPS (Power Supply Unit)** – Converts AC power into DC power and supplies electricity to all components.
7. **CPU Cooler/Fan** – Keeps the processor cool and prevents overheating.
8. **Cabinet Fan** – Improves airflow inside the cabinet and keeps the system cool.
9. **Graphics Card (GPU)** – Processes graphics and improves gaming, video editing, and display performance.
10. **Optical Drive (DVD Drive)** – Reads and writes CDs or DVDs (available in some desktop PCs).
11. **SATA and Power Cables** – Connect storage devices to the motherboard and provide power to all components.
12. **Ethernet Port**—Provides internet through Cable.

**Q2. Take clear photos or draw labelled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

Step 1: Unplug the Power Cable



Caption:

First, switch off the computer and unplug the power cable from the SMPS (Power Supply Unit). This prevents electric shock and protects the hardware.

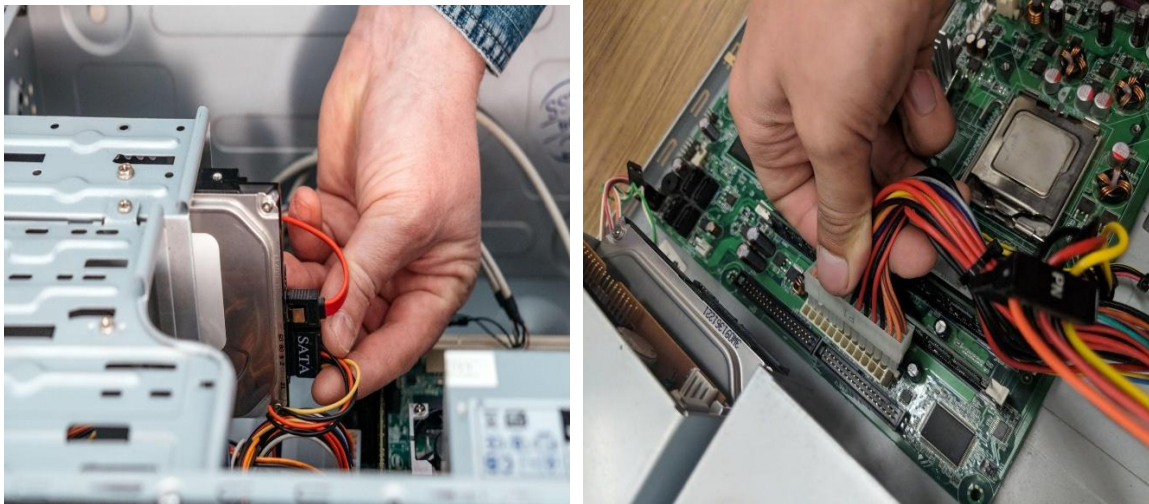
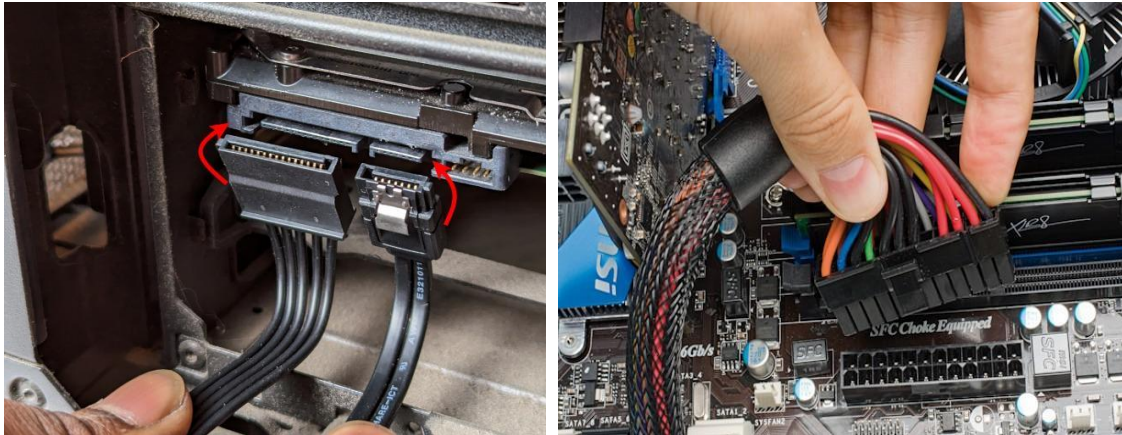
Step 2: Remove the Side Panel



Caption:

Unscrew the side panel and slide it backward to open the computer cabinet. Now all internal components are visible.

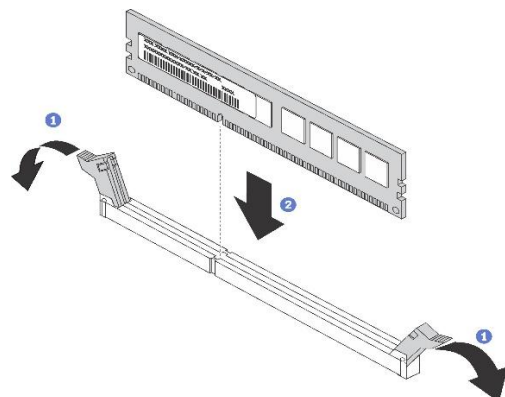
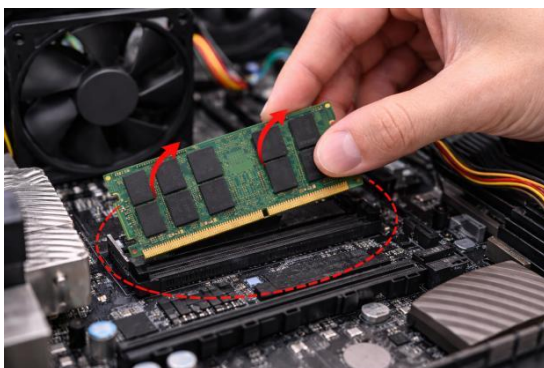
Step 3: Disconnect Internal Cables

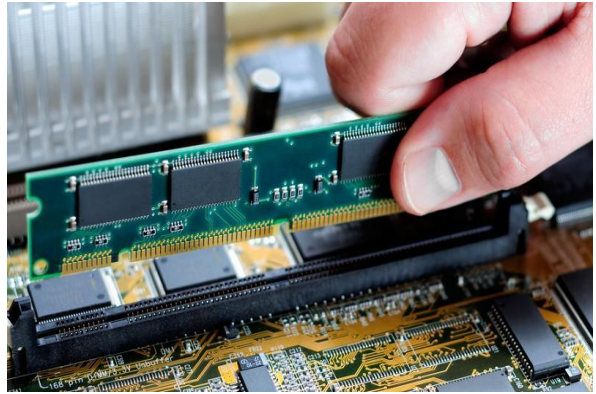
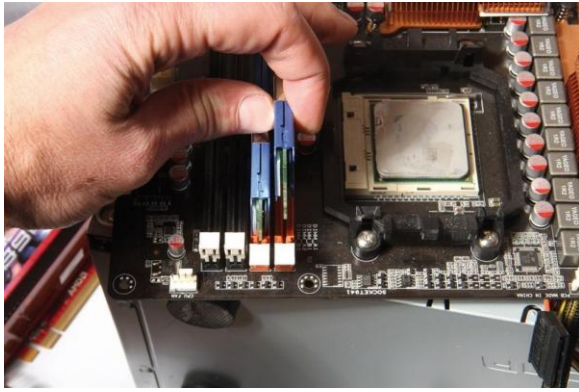


Caption:

Carefully disconnect the SATA data cable, SATA power cable, and motherboard power connectors before removing components.

Step 4: Remove the RAM

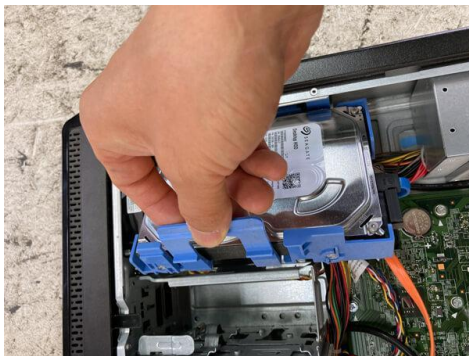




Caption:

Press the clips on both sides of the RAM slot. The RAM module pops up and can be removed carefully.

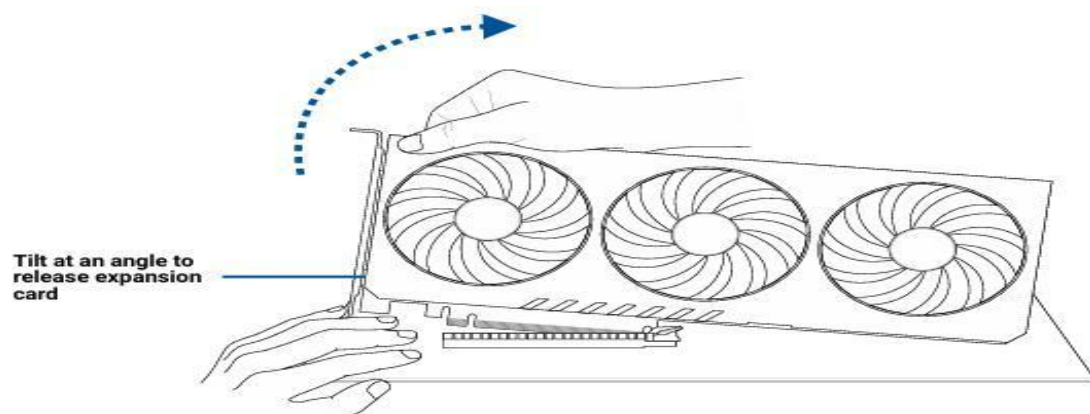
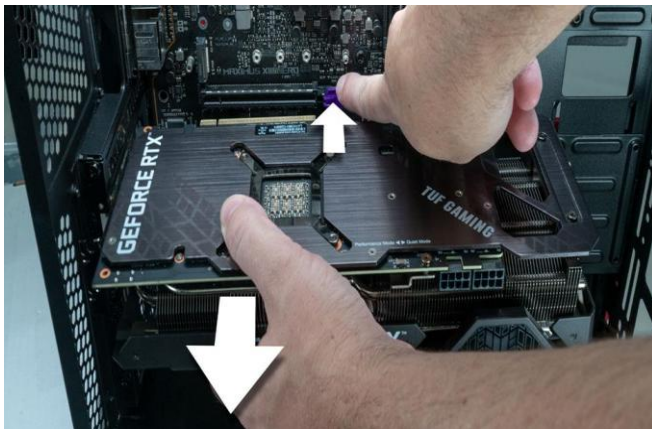
Step 5: Remove the HDD/SSD



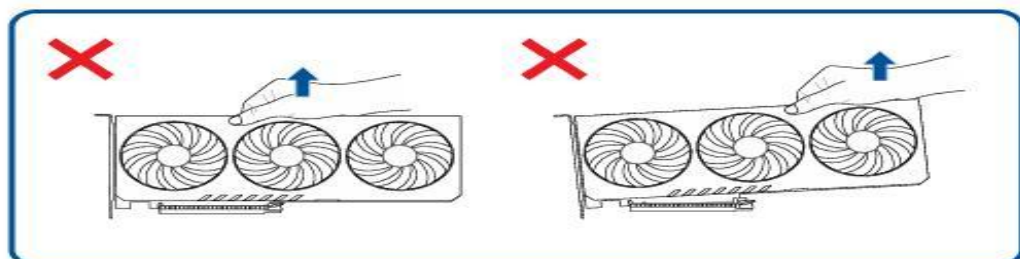
Caption:

Remove the screws holding the HDD or SSD, disconnect the cables, and gently slide the drive out of the drive bay.

Step 6: Remove the Graphics Card (If installed)



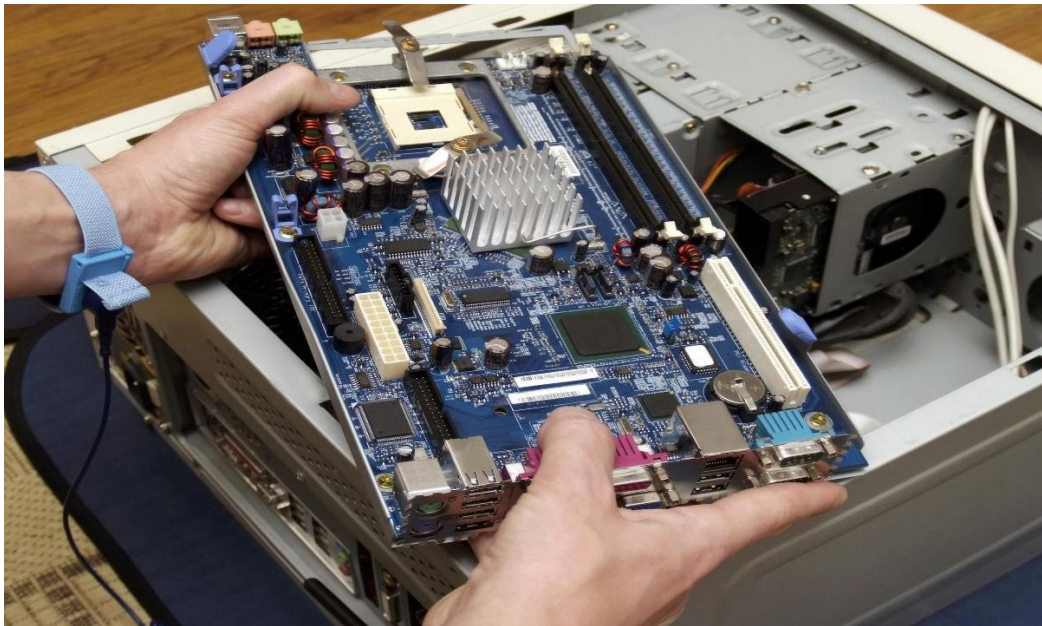
Tilt at an angle to release expansion card



Caption:

Unscrew the graphics card, press the PCIe slot latch, and carefully pull the graphics card out.

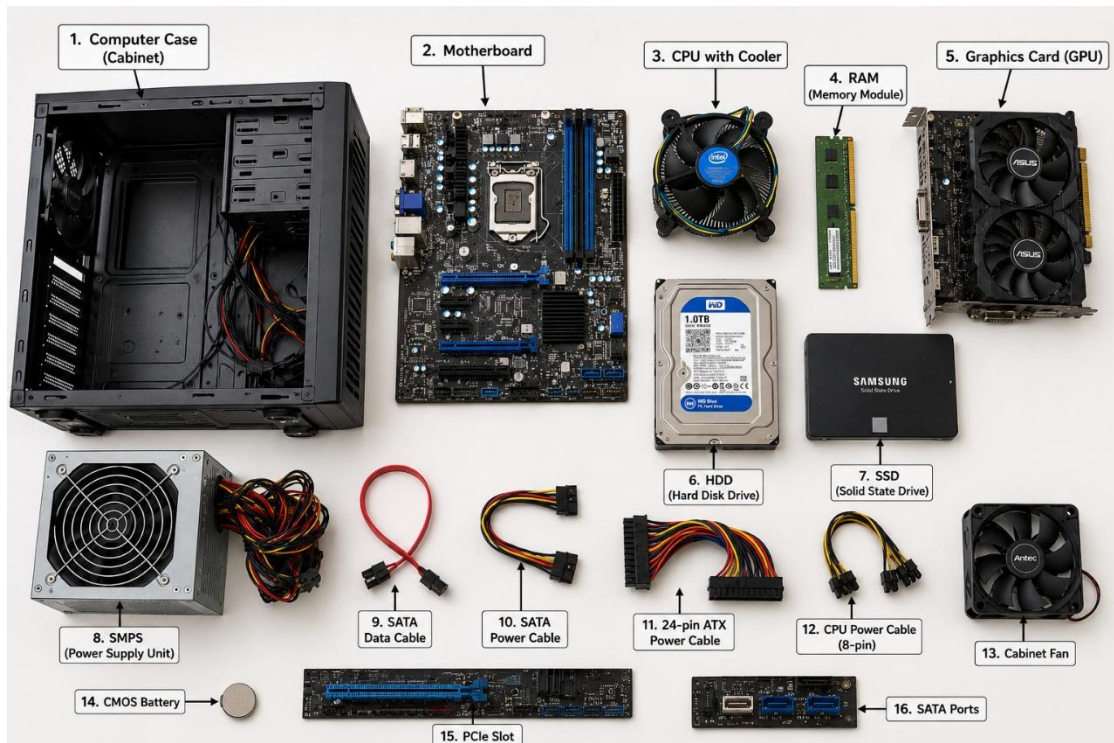
Step 7: Remove the Motherboard



Caption:

Remove all motherboard screws and lift the motherboard out carefully while holding it by the edges.

Step 8: Components After Disassembly



Caption:

Arrange all removed components neatly, such as the motherboard, RAM, HDD/SSD, graphics card, SMPS, and cables, for inspection or reassembly.

Q3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Reassembling the Desktop PC

➤ Step 1: Install the Motherboard

Put the motherboard inside the cabinet and tighten the screws.

Why: It is the main board where all parts connect.

➤ Step 2: Install the CPU

Place the CPU in the CPU socket and lock it.

Why: The CPU is the brain of the computer.

➤ Step 3: Install the CPU Cooler

Fix the CPU cooler on the CPU and connect its fan cable.

Why: It keeps the CPU cool.

➤ Step 4: Install the RAM

Insert the RAM into the RAM slot until it clicks.

Why: RAM is needed to run programs.

➤ Step 5: Install the SMPS

Fix the SMPS inside the cabinet with screws.

Why: It supplies power to the computer.

➤ Step 6: Install the HDD/SSD

Place the HDD or SSD in the drive bay and tighten the screws.

Why: It stores Windows and files.

➤ Step 7: Connect the Cables

Connect:

- SATA data cable
- SATA power cable
- 24-pin motherboard power cable
- 8-pin CPU power cable

Why: These cables give power and transfer data.

➤ **Step 8: Install the Graphics Card (If Available)**

Insert the graphics card into the PCIe slot and tighten the screw.

Why: It improves display and gaming performance.

➤ **Step 9: Connect Front Panel Cables**

Connect the Power Switch, Reset Switch, USB, Audio, and Fan cables.

Why: These make the front buttons and ports work.

➤ **Step 10: Close the Cabinet**

Put the side panel back and tighten the screws.

Why: It protects the computer parts.

➤ **Step 11: Connect External Devices**

Connect the monitor, keyboard, mouse, and power cable.

Why: These are needed to use the computer.

➤ **Step 12: Turn On the PC**

Press the power button and check if the computer starts.

Why: This confirms that all parts are connected correctly.

Q4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

➤ **Safety Precautions**

☐ **Turn off the PC**

Why: Prevents damage to the computer.

☐ **Unplug the power cable**

Why: Protects you from electric shock.

☐ **Press the power button for 5–10 seconds**

Why: Removes any remaining power from the PC.

☐ **Work on a clean and dry table**

Why: Keeps components safe from dust and water.

☐ **Touch a metal surface or wear an anti-static wrist strap**

Why: Prevents static electricity from damaging components.

☐ **Handle components by the edges**

Why: Avoids damage to chips and circuits.

☐ **Keep screws in a small container**

Why: Prevents screws from getting lost.

➤ **Tools Needed**

☐ **Phillips Screwdriver**

Why: Used to remove and tighten screws.

☐ **Anti-static Wrist Strap**

Why: Protects components from static electricity.

☐ **Small Container for Screws**

Why: Keeps screws organized.

☐ **Soft Brush**

Why: Removes dust from components.

☐ **Microfiber Cloth**

Why: Cleans parts without scratching them.

☐ **Cable Ties**

Why: Keeps cables neat and improves airflow.

☐ **Flashlight (Optional)**

Why: Helps you see small connectors inside the cabinet.